Smart City Healthcare Network Design

CSE421 | Section 02 | Project Report

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1. Project Overview

The project designs a network in Cisco Packet Tracer for six connected healthcare units: Main Hospital (MH), Diagnostic Center (DC), Emergency Response Unit (ERU), Research Institute (RI), Pharmacy Hub (PH), and Remote Health Clinic (RHC). The design accommodates **810 planned devices**, while using only representative PCs in the simulation. It provides inter-unit communication, centralized DNS and web access, email exchange, automatic addressing, and backup connectivity.

Cisco 2911 routers, 2960 switches, and Server-PT devices are used. The base IPv4 block is `1.1.0.0/16`, following the assignment's ID-based rule for the third group member (digits `01`). This is a classroom simulation range rather than a recommended private-address block for production.

2. Network Topology

MH, RI, and ERU form a triangular backbone. MH also connects through a shared switch to DC, PH, and RHC. An additional direct RHC–ERU link supports backup routing for the remote clinic.

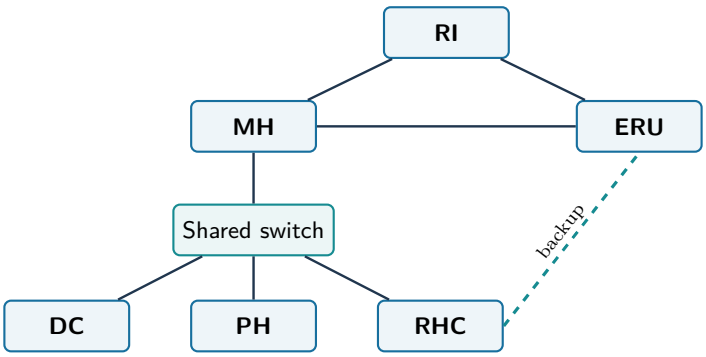


Figure 1: Logical network structure. Solid lines show ordinary connections; the dashed link connects RHC to ERU for backup. Each healthcare unit also has its own LAN switch.

3. VLSM Addressing Plan

Variable Length Subnet Masking (VLSM) assigns each unit the smallest subnet that supports its device requirement. For example, MH requires 300 addresses, so it uses a `/23` subnet with 510 usable addresses instead of a `/24`, which provides only 254.

Table 1: Healthcare LANs and gateway allocation

Unit	Devices	Usable	Network	Gateway
MH	300	510	`1.1.0.0/23`	`1.1.0.1`
DC	150	254	`1.1.2.0/24`	`1.1.2.1`
RI	120	126	`1.1.3.0/25`	`1.1.3.1`
PH	100	126	`1.1.3.128/25`	`1.1.3.129`
ERU	80	126	`1.1.4.0/25`	`1.1.4.1`
RHC	60	62	`1.1.4.128/26`	`1.1.4.129`
Total	**810**			

Backbone subnets: MH–RI `1.1.5.0/30`; MH–ERU `1.1.5.4/30`; ERU–RI `1.1.5.8/30`; shared transit `1.1.5.16/29`; and RHC–ERU backup `1.1.5.24/30`.

4. Network Services

The project uses both static addressing and DHCP. Main Hospital and Research Institute have statically addressed hosts and servers, while DC and ERU receive DHCP leases from their routers. PH and RHC have dedicated DHCP servers.

Table 2: Essential service configuration

Service	Location / IP	Configuration
DHCP	DC, ERU routers	Local router DHCP pools
DHCP	PH: `1.1.3.140`	Server leases begin at `1.1.3.160`
DHCP	RHC: `1.1.4.140`	Server leases begin at `1.1.4.160`
DNS	MH: `1.1.0.10`	Resolves healthcare service domains
Web	MH: `1.1.0.20`	`www.smartcityhealth.local`
Email	MH: `1.1.0.30`	`admin@mh.com`
Email	RI: `1.1.3.30`	`admin@ri.com`

The centralized DNS server maps the web domain to `1.1.0.20` and the two email domains to their respective servers. The MH and RI email servers use SMTP and POP3 in Packet Tracer to support sending and receiving messages between the units.

5. Routing and Redundancy

RIPv2 is configured on the routers using `version 2`, `no auto-summary`, and the classful RIP network statement `network 1.0.0.0`. RIPv2 exchanges routes for the VLSM subnets. Four specific static-routing cases complement the dynamic routes:

Table 3: Routing requirements implemented in the design

Router	Route type	Destination and route selection
MH	Next-hop static	DC `1.1.2.0/24` via `1.1.5.18`
ERU	Exit-interface static	MH `1.1.0.0/23` via `G0/1`
PH	Recursive static	DC `1.1.2.0/24` via `1.1.5.2`
RHC	Floating static	MH `1.1.0.0/23` via ERU `1.1.5.26`, AD 130

The RHC floating static route has an administrative distance of **130**, greater than RIPv2's **120**. Therefore, it is intended to become active only when a preferred RIP route to MH is removed. The PH route is recursive because its specified next hop is not directly connected; the actual forwarding path depends on the routing table.

6. Verification and Conclusion

The supplied walkthrough identifies the following Packet Tracer checks. These are **planned/expected verification results**, not independent measurements or screenshots supplied with this report.

Table 4: Principal validation checks

Test	Expected evidence
Connectivity	`ping` between units; configured interfaces show `up/up`.
Routing	`show ip route` displays connected, static, and RIP-learned routes.
DNS / HTTP	`www.smartcityhealth.local` resolves and loads from PCs.
Email	MH sends to `admin@ri.com`; RI receives and replies.
MH–RI failure	Shut MH `G0/1`; RI reaches MH through ERU after reconvergence.
RHC backup	Shut RHC `G0/1`; check `show ip route`, `tracert`, and connectivity via ERU.

Conclusion. The proposed design connects all six healthcare units through appropriately sized subnets, centralized services, and a combination of dynamic and static routing. The MH–RI–ERU triangle and the dedicated RHC–ERU link provide alternative paths in the simulated network. Successful DNS, web, email, and failover tests should be supported with Packet Tracer outputs before reporting them as experimentally verified.